American Computer Science League

2023-2024 ● Contest 1: Short Problems Solutions ● Intermediate Division

1. Computer Number Systems

$2023_{10} = 3747_8$
3747, 3750, …, 3757, 3760, … 3767, 3771, 3772, 3773 is a palindrome.
$3773_8 = 3 * 8\text{^}3 + 7 * 8\text{^}2 + 7 * 8 + 3 = 3 * 512 + 7 * 64 + 56 + 3 = 2043_{10}$

D. 2043

2. Computer Number Systems

$F43A6C3091_{16} = 1111\ 0100\ 0011\ 1010\ 0110\ 1100\ 0011\ 0000\ 1001\ 0001_2$
$= 1\ 111\ 010\ 000\ 111\ 010\ 011\ 011\ 000\ 011\ 000\ 010\ 010\ 001_2$
$= 1\ \ 7\ \ 2\ \ 0\ \ 7\ \ 2\ \ 3\ \ 3\ \ 0\ \ 3\ \ 0\ \ 2\ \ 2\ \ 1_8$
There are 3 0's, 2 1's, 4 2's, 3 3's, 0 4's, 0 5's, 0 6's, and 2 7's. A 2 occurs most.

C. 2

3. Recursive Functions

$f(17, 11) = f(11 - 1, 17 - 2) + 1 = f(10, 15) + 1 = 15 + 1 = 16$
$f(10, 15) = f(10 - 1, 15 - 2) + 3 = f(9, 13) + 3 = 12 + 3 = 15$
$f(9, 13) = f(9 - 1, 13 - 2) + 3 = f(8, 11) + 3 = 9 + 3 = 12$
$f(8, 11) = f(8 - 1, 11 - 2) + 3 = f(7, 9) + 3 = 6 + 3 = 9$
$f(7, 9) = f(7 - 1, 9 - 2) + 3 = f(6, 7) + 3 = 3 + 3 = 6$
$f(6, 7) = f(6 - 1, 7 - 2) + 3 = f(5, 5) + 3 = 0 + 3 = 3$
$f(5, 5) = 5 - 5 = 0$

E. 16

4. Recursive Functions

$f(24) = 2 * f(24 - 4) = 2 * f(20) = 2 * 60 = 120$
$f(20) = 2 * f(20 - 4) = 2 * f(16) = 2 * 30 = 60$
$f(16) = f(16 + 1) - 1 = f(17) - 1 = 31 - 1 = 30$
$f(17) = f(17 + 1) - 1 = f(18) - 1 = 32 - 1 = 31$
$f(18) = 2 * f(18 - 4) = 2 * f(14) = 2 * 16 = 32$
$f(14) = 14 + 2 = 16$

D. 120

5. What Does This Program Do? (Branching) Given the rules to follow, only the amount over the amount specified is given the greatest percent. The other categories are paid 3% of the second $5000 and 1% of the first $5000 which is $50. The results are: a) the amount over a certain threshold is never used b) this works correctly though not as efficiently designed c) this is not as well written, but also works d) this works correctly by using the opposite order for the if…else if…else Therefore, the answer is b, c, d.	A. b, c, d